

TSPI — Phase 3 Expert Feedback: Deep Analysis vs. Our Implementation

Analysis of the domain experts' Phase-3 feedback against the `tspi_ai_brain` engine as currently built.

Source documents analysed (all read in full)

#	File (in <code>docs/</code>)	Pages	Contents
[A]	<code>tspi comment for phase3 developer_260713_104231.pdf</code>	36	Part 1 (Clinical Reasoning Core) + Phase-3 report review (13 points) + canonical 39-Axis Master Dictionary + Priority 0/1 list
[B]	<code>tspi comment for phase3 developer .pdf</code>	55	Pages 1-47 are identical to [A] , plus NEW pages 48-55: "Part 6 of 6 — The Fundamental Philosophy and Intelligence Architecture of TSPI-AI" (see §6b)
[C]	<code>TSPI AI Engine — Domain Expert Answers for the Dev_260713_095402.pdf</code>	22	Answers to our 7 remaining questions (marker→axis, thresholds, NSS formula, Axes 37-39, learning rule, approver, language)
[D]	<code>tspi developer answer.pdf</code>	3	Earlier module-registry answers (1 module = 1 product, H-Code, etc.)

[B] supersedes [A] (it contains everything in [A] plus Part 6). Both are cited below because [A] was the first received. **[C] contradicts [A]/[B]** on Axes 35-39 — see Critical #1.

0. Scope note — read this first

The review examines **three Phase-3 report documents**: *TSPI Clinic — Clinical Intelligence Platform Physician Edition (48 pp)*, *Patient Edition (25 pp)*, and *Multi-Omics Edition (51 pp)*. **Our engine did not produce those reports**. They contain things our engine cannot emit — invented module names ("Iron Recovery Module"), `AXIS_7 = 85/100` generic labels, projected omics values, and default 50/100 scores.

So the review is of a **different artifact**. However, the **principles** it lays down are binding on TSPI as a whole, and most of them expose **real gaps in our design**. This document separates:

- criticisms that **do not apply to us** (we already comply), and
- criticisms that **do apply** and require work.

Two findings are critical and block clinical use. Both are ours.

1. ♦ CRITICAL #1 — Two expert PDFs contradict each other on Axes 35-39

The two conflicting source files (both dated 260713, ~1 hour apart):

- **Source [C]** — `TSPI AI Engine — Domain Expert Answers for the Dev_260713_095402.pdf` (22 pp), **Q4A "Where do Axes 37-39 belong?"** (pages 7-9)
- **Source [A]/[B]** — `tspi comment for phase3 developer_260713_104231.pdf` (36 pp) and `tspi comment for phase3 developer .pdf` (55 pp), section **"3. Master Dictionary that the Developer**

must LOCK across all 39 axes [Thai]" — Domain 12 — Oncology & System-Level Regulation (pp. 29–30 / 39)

They give **incompatible** definitions of the top of the axis list:

Axis	[C] Domain Expert Answers PDF (Q4A)	[A]/[B] Phase-3 Comment PDF (Master Dictionary)	Our engine today
35	—	Visceral Organ Functional Reserve	Organ Resilience & Functional Reserve
36	—	Local Organ & Tissue Interface	Oncology
37	Protein Quality Control System	Oncologic Cell Fate	Protein Quality Control System
38	Protein Clearance & Proteotoxic Stress	Tumor Microenvironment & Metastatic Dynamics	Protein Clearance & Proteotoxic Stress
39	Cognitive-Emotional Loop Axis (CELA)	Genomic Regulation Homeostasis	CELA
Domain 12	"Proteostasis, Cellular Integrity and Conscious Regulation "	" Oncology & System-Level Regulation"	Proteostasis & Cellular Integrity

Note both Phase-3 files ([A] 36 pp and [B] 55 pp) carry the **same** Master Dictionary, so this is a genuine disagreement between documents — **not** a transcription or OCR error on our side.

[A]/[B] goes further and states **proteostasis/autophagy lives at Axis 10** (*Proteostasis–Autophagy–Lysosome*), explicitly listing "*proteostasis/autophagy — because it belongs to Axis 10 [Thai]*" under "**must NOT infer Axis 37 from**". That is irreconcilable with [C], which puts *Protein Quality Control* at Axis 37.

Our dictionary matches 34/39 of the Phase-3 master. The **only** divergences are axes **35–39** — precisely the disputed region. Note the good news: the "semantic drift" the expert is angry about (Axis 7 = inflammation/haematopoiesis, Axis 10 = microbiome, etc.) is **not present in our data** — our Axis 7 = *Nutrient Sensing* and Axis 10 = *Proteostasis–Autophagy–Lysosome* already match the master.

ACTION REQUIRED — cannot be resolved by us. The client must declare **which of the two axis dictionaries is canonical**. Everything downstream (axis scoring, module mapping, reports) depends on it. Until then, axes 35–39 must be treated as unstable.

2. ♦ CRITICAL #2 — Our module→axis map uses the LEGACY axis numbering

The expert states plainly: "*Do not take the Axis Mapping column from the product file directly into production. Every product must be remapped to the latest Axis Master first.*" The product sheets use an **old numbering** that is offset in the middle of the range:

Product file (legacy)	Master (correct)
16 = Liver detoxification	15 = Detoxification Capacity
17 = Digestive function	16 = Digestive-Absorptive Function
18 = Microbiome ecology	17 = Microbiome Ecology
19 = Gut barrier	18 = Gut Barrier & Mucosal Integrity
21 = Endothelial function	20 = Endothelial Function
25 = Tissue repair	24 = Wound Healing
27 = Metabolic balance	27 = Glycation / Carbonyl Stress (<i>meaning changed!</i>)

We built `axis_module_official.json` straight from that column. Verified: our registry maps **H-001 Curcuma** → **A1, A9, A16, A17, A18, A19, A21, A25, A27** — the legacy numbers exactly.

Consequence: every module→axis link in our registry is **systematically wrong** in the A15–A27 band (roughly an off-by-one), so module selection is clinically incorrect. This also silently contradicts the earlier owner ruling ("*ignore `target_axes_raw` in the legacy product sheets*") which we did not honour when compiling the draft.

ACTION REQUIRED. Build a **legacy→master axis remap table**, apply it in `build_registry.py`, and add a validator rule that refuses to compile un-remapped mappings. Blocked on Critical #1 for axes 35–39.

3. The philosophical demand: symptoms are primary evidence

This is the heart of Part 1 of the review, and it is **the biggest design gap we have**.

What they demand:

- Patient **symptoms are biological evidence**, not secondary text. An axis must be assessable **without** specialist biomarkers.
- **Forbidden:** "no abnormal lab → axis normal"; "no biomarker → axis cannot be assessed"; "normal test → normal network".
- A **normal test narrows the differential, it does not erase symptoms** (normal gastroscopy does not exclude motility/acid/enteric/microbiome dysfunction).
- Required: a **Clinical Phenotype Engine** with structured symptoms (onset, duration, frequency, severity, trigger, relieving factor, meal/sleep/stress/bowel relation, functional impact, associated symptoms, red flags), **symptom clusters**, a **Symptom→Axis** and **Symptom→Network** dictionary, **adaptive follow-up questioning**, and **probabilistic differential network mapping** (never one network from one symptom).
- **Red-flag screening comes first:** Symptoms → Red-flag screening → Standard medical differential → Functional network mapping → Axis integration → Module consideration.
- Evidence layers 1–5 (phenotype → exam → routine labs → imaging → omics). **Layer 5 must not be required** to analyse.

What we have: `PatientInput.symptoms` is a **plain free-text string**. Axis scoring is **lab-driven**. There is no structured phenotype, no symptom→axis dictionary, no clusters, no red-flag layer, no adaptive questioning.

This is exactly the failure we already diagnosed ourselves in `TSPI_Report_Comparison_Diagnosis.md` — a symptoms-only case produced a nearly empty report. The expert has now told us *why* that is philosophically wrong, not just operationally thin.

Verdict: major gap. This is the single largest piece of work implied by the feedback.

4. Full gap matrix — demand vs. our engine

#	Expert requirement	Our engine today	Verdict
1	Axis scores computed by a structured engine, not the LLM	Deterministic pipeline; LLM only writes prose	✓ Comply
2	Only official registry modules — AI may not invent module names	Registry-backed; unresolved modules flagged	✓ Comply (just built)
3	Physician approval before patient release	/validate gate; deliverable=false until approved	✓ Comply
4	No default 50/100 — use NOT_ASSESSED + INSUFFICIENT_DATA	We never emit a default 50	✓ Comply (but see #6)
5	Single versioned severity band from backend, not per-report	One dosing/severity config; version-stamped	✓ Comply
6	Explicit NOT_ASSESSED / reason for unmeasured axes	Unscored axes are simply absent — silent, not explicit	! Gap (small)
7	Separate 4 scores : symptom burden, axis probability, evidence confidence, data completeness	Single axis score only	✗ Gap (large) — <i>also required by the NSS formula in the other PDF</i>
8	evidence_status on every statement (MEASURED / DERIVED / CLINICAL_INFERENCE / HYPOTHESIS / NOT_AVAILABLE); HYPOTHESIS must never raise a score	Not modelled	✗ Gap
9	Symptoms as core evidence + structured phenotype + clusters	Free-text string; lab-driven	✗ Gap (largest)
10	Symptom→Axis and Symptom→Network dictionaries	None	✗ Gap
11	Red-flag screening before network analysis	None (module safety only)	✗ Gap (safety-critical)
12	Negative-test interpretation rules	None	✗ Gap
13	Adaptive clinical questioning	None	✗ Gap
14	180 Networks layer; 39↔180 fixed table; network propagation	Absent — we jump axes → modules. Graph engine works over axes only	✗ Gap (architectural)
15	12 Domains + 3 Keys rollup in the output; 9-Step restoration map	Domains/keys in DB; 9-step locator exists ; not surfaced as Keys/Domain dashboards	! Partial
16	Module match formula (30% axis / 25% network / 15% evidence grade / 10% phenotype / 10% safety / 5% step / 5% history) + deductions + reason_not_selected	Simple axis match, no scoring, no rejection reasons	✗ Gap
17	Evidence grades A/B/C/D linked to research	None	✗ Gap
18	Bilingual (Thai default patient-facing, English canonical)	English only	✗ Gap
19	Versioning of axis / network / product mapping	registry_version + framework_version stamped	✓ Comply (extend to network/axis)
20	Learning loop: patient-specific bounded + population propose-only	Global auto-recalibration of axis weights	✗ Non-compliant (per the other PDF)
21	Digital twin / simulation , TSPI knowledge graph	None	○ Future (P2)
22	Correct axis dictionary	34/39 match; 35-39 disputed	◆ Blocked on Critical #1

#	Expert requirement	Our engine today	Verdict
23	Remapped product→axis mapping	Legacy numbering	◆ Critical #2

5. What we already get right (do not rebuild)

The expert scored the reviewed system *Software architecture 90/100, Future scalability 98/100* — and the things it praised are things our engine already enforces structurally:

- Deterministic, auditable scoring engine — **the LLM never assigns an axis score.**
- Modules resolve to the **official catalogue** or are flagged; nothing invented.
- **Physician-in-the-loop** gate; nothing reaches a patient un-approved.
- De-identification; append-only audit; consent gating.
- Data-driven design — axis/module/dosing all load from versioned files, so fixing the dictionary and the remap is a **data change, not a rewrite.**

The expert's own conclusion applies to us too: "*What must be done is not to rebuild the software, but to freeze the new master ontology and make every calculation flow from real data → axis.*"

6. Recommended priority plan

P0 — Blocking (no clinical use until done)

1. **Resolve the axis-dictionary conflict** (Critical #1) — client decision required.
2. **Build the legacy→master axis remap table**, apply in `build_registry.py`, and add a validator rule that fails on un-remapped input (Critical #2).
3. **Add confidence + data_completeness + NOT_ASSESSED** to axis scoring (also unlocks the official NSS formula, which needs `Confidence_i`).
4. **Add evidence_status** to every derived statement; forbid HYPOTHESIS from affecting any score.
5. **Red-flag screening layer** before network/module reasoning (safety-critical).
6. **Fix the learning loop** to bounded patient-specific + propose-only population learning.

P1 — Makes it genuinely TSPI

1. **Clinical Phenotype Engine**: structured symptoms + clusters + Symptom→Axis dictionary (turns symptoms into first-class evidence — fixes the thin-report problem at the root).
2. **180 Networks layer**: network master table, 39↔180 mapping, network propagation.
3. **Module match scoring formula** + `reason_not_selected`.
4. **3 Keys / 12 Domains / 9-Step** rollups in the report output.
5. **Negative-test interpretation rules**; adaptive follow-up questioning.
6. **Bilingual** output (Thai default, English canonical).

P2 — Differentiators

1. Evidence grades A–D; TSPI knowledge graph; digital-twin simulation (module A vs B → predicted 3-month trajectory).

6b. ★ NEW — "Part 6 of 6: The Fundamental Philosophy" (from the 55-page edition)

Source **[B]** (`tspi_comment_for_phase3_developer.pdf`, 55 pp) contains pages 1–47 identical to **[A]** (the 36-page file), **plus a new closing section (pp. 48–55)** that the expert calls "*the final and most important*

axis." It is not a feature request — it defines **how every algorithm inside TSPI-AI must reason**.
"Without this philosophy, TSPI-AI will become another disease-oriented clinical AI. With it, TSPI-AI becomes a Biological Intelligence Platform."

The Final Core Principle (to become the FIRST design principle of the whole engine)

"In Fundamental Medicine, symptoms and diseases are not the primary targets of treatment. They are biological expressions of dysregulated living biological networks. Therefore, TSPI-AI reasons **from biological networks toward symptoms and diseases — not from diseases toward biological networks**."

The causal chain (note the direction)

Biological Network Dysregulation → Symptoms → Laboratory Changes → Structural Changes → Disease

Disease is a **consequence, not the starting point**. Labs are **downstream events**; imaging/structural change is **later still**. Hence: *normal labs ≠ normal biology*, and *abnormal labs ≠ the primary driver*.

Ten principles that bind our engine

1. **Disease is not the treatment target** — diseases (Diabetes, GERD, Migraine) are *clinical labels*, useful for communication but **not the reasoning units**. The reasoning units are **living networks**.
2. **Symptoms are biological language** — not noise, not subjective, not secondary. (Constipation may express enteric-NS, colonic transit, microbiome, autonomic, thyroid, mitochondrial, hydration...)
3. **Every symptom → multiple candidate networks → produce a differential network model**, never a single diagnosis.
4. **Evidence ≠ Truth**. Absence of microbiome sequencing does not mean the microbiome is normal — it means *insufficient direct evidence*.
5. **Every conclusion must always expose FOUR dimensions** (this extends what we had):
 - **Evidence Level**
 - **Confidence Level**
 - **Data Completeness**
 - **Biological Uncertainty** ← *new fourth dimension*
6. **Treatment = Network Harmonization** — the vocabulary must change from *treat / correct / reduce / improve* to **Network Harmonization, Adaptive Compensation, Biological Resilience, Network Plasticity**.
7. **Reasoning must be dynamic** — every new datum **reconstructs** the patient's biological model. New evidence may *confirm, refine, redirect, override or contradict* prior hypotheses. The AI must **not preserve a static diagnosis**.
8. **The objective is not normal lab values** — it is restored **resilience, adaptive capacity and physiological harmony** of the whole organism.
9. **TSPI-AI is a Biological Intelligence System**, not a lab reader or an auto-prescriber.
10. **Mission:** "*the world's first Biological Intelligence Operating System based on the philosophy of Fundamental Medicine*" — explicitly **not** "another clinical decision support system."

The Adaptive Biological Intelligence Loop (new core architecture — distinct from our learning loop)

Patient → Symptoms & History → Clinical Phenotype → 39 Biological Axes
→ 180 Living Biological Networks → Root Driver Analysis
→ Network Harmonization Plan → Clinical Outcome → Network Response
→ Biological Model Update → Better Biological Understanding → Next Clinical Decision

The decisive sentence: "What the AI learns is not 'disease' — it is **the behaviour of biological networks**." This is stated as the single most important difference between TSPI-AI and generic clinical AI.

What this changes for us (beyond §4)

Part-6 demand	Our engine today	Verdict
Reason network → symptom/disease , not the reverse	We reason lab/finding → signal → axis → module . Serviceable as <i>inference</i> , but the engine has no network layer to reason from, and the report is organised around findings/conditions	✗ Architectural gap
Diseases are labels, not reasoning units	<code>conditions[]</code> is an input that only feeds safety checks — we never reason <i>from</i> them, which is accidentally correct, but we also never reason from networks	! Partial
4th dimension: Biological Uncertainty on every conclusion	Not modelled (we have none of the four)	✗ Gap — <i>extends §4 item 7</i>
Network Harmonization vocabulary	Report language is treat/correct/reduce	✗ Gap (report_composer prompt + templates)
Dynamic model reconstruction — patient's biological model is rebuilt on every new datum	Each <code>/report</code> is a one-shot, immutable artifact. <code>/temporal</code> only trends raw markers	✗ Gap — needs a persistent, versioned <i>patient biological model</i>
Adaptive Biological Intelligence Loop — learn network behaviour	Phase 4 learns axis weights from marker deltas . Wrong learning target <i>and</i> (per the other PDF) wrong scope (global auto-update)	✗ Redesign required
Root Driver Analysis at network level	<code>graph_engine</code> does PageRank root-cause over axes — the closest thing we have, but not over the 180 networks	! Partial — good foundation, wrong layer

Net effect: Part 6 promotes several items from "nice to have" to **architectural**. In particular the **180-Networks layer stops being optional** — it is the layer the entire system is supposed to *reason in*. Our axis-level graph engine is the right machinery pointed at the wrong layer.

7. Questions the client must answer before P0 can start

1. **Which axis dictionary is canonical** — [C] TSPI AI Engine – Domain Expert Answers for the `Dev_260713_095402.pdf` (A37 = Protein Quality Control, A38 = Protein Clearance, A39 = CELA), **or** [A]/[B] `tspi comment for phase3 developer ...pdf` Master Dictionary (A37 = Oncologic Cell Fate, A38 = Tumor Microenvironment, A39 = Genomic Regulation Homeostasis)? **They cannot both be right**, and every downstream calculation depends on the answer.
2. Is there an **official legacy→master axis conversion table**, or should we derive one and have it clinically reviewed?
3. The **180 Networks master table** (network_code, name, axis_code, type, upstream/downstream) — does it exist in a loadable form?
4. **3 Keys ordering:** the master lists *Biological Dynamics* before *Metabolic Energy*, while older data uses Metabolic Energy as Key 1. Confirm one standard (and keep `key_code` separate from display order, as the expert advises).